LESSON PLAN · SCRATCH (GR 3–5) ADVANCED: ARCADE ENGINEERS

Lesson 12: Capstone: Arcade Engineer

Module 4 | 90 minutes | Capstone project | Grades 3–5 · ages 9–11

Objective: Design, build, test and present an original arcade game that uses clones, variables, a custom block or list, and game states.

Success criteria

- My game uses clones or lists, a custom block and game states.
- It resets and ends properly.
- I playtested, improved and presented it.

Key vocabulary

- **game design**: A plan of how your game works.
- **playtest**: Watching others play to find problems.
- **iteration**: Improving in small cycles.

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)
- Printed project brief

Timed lesson plan (90 min)

Time	Stage	What happens
10 min	**Session 1 · Design**	Game design sheet; teacher signs off.
30 min	**Session 1 · Build core**	Player and main rule; test as you go.
5 min	**Session 1 · Save**	Save progress; note the next feature.
20 min	**Session 2 · States & blocks**	Menu, levels/lives, game over; refactor into a My Block.
10 min	**Session 2 · Playtest**	Two testers each; improve one thing.
15 min	**Session 2 · Submit & present**	Reflection, submit, 1-minute presentations.

Unplugged warm-up

Design pitch (5 min): pairs pitch their game idea in 30 seconds each.

Teaching the idea

The capstone assesses the whole course: clones, velocity, lists, procedures, broadcasts and state variables, plus the design–build–test cycle. Run over two sessions with a design checkpoint.

Analogy: An engineer's demo day: plan, build, test, present.

Live demo: Show a minimal playable version, then add one feature live and test it.

Questions to ask

Ask	Listen for
What is your smallest playable version?	Player + one rule.
Where did you use a My Block?	To remove repeated code.
What did testers change your mind about?	Specific feedback.

Differentiation

Support
Allow remixing the Starfall Catcher or Platformer project with added states and a block.

Stretch
Multiple levels, a high-score list, sound design.

Common misconceptions

Misconception	How to fix it
Build everything before testing	Test after each feature.
More features = better game	A polished small game beats a buggy big one.

Assessment

Project rubric (also on the project brief):

Criterion	1 · Beginning	2 · Developing	3 · Secure	4 · Excellent
Design & originality	No plan	Copied idea	Original idea with a plan	Creative idea; plan followed and updated
Advanced techniques	None	One of clones/lists/My Block	Two used correctly	All three used well

Game states	None	Start or end only	Menu, play and game over	Levels or lives too
Code quality	Hard to follow	Works but repetitive	Tidy with a My Block and resets	Clear, well-named and efficient
Testing & presentation	Not tested	Tested alone	Two playtesters; improvement explained	Clear presentation of code and changes

Exit ticket
One thing you would add in version 2.

Homework
Share your game with family and collect one piece of feedback.

Answer key — Kitabash activities

1. Which tool would you use? (perform · categorize)

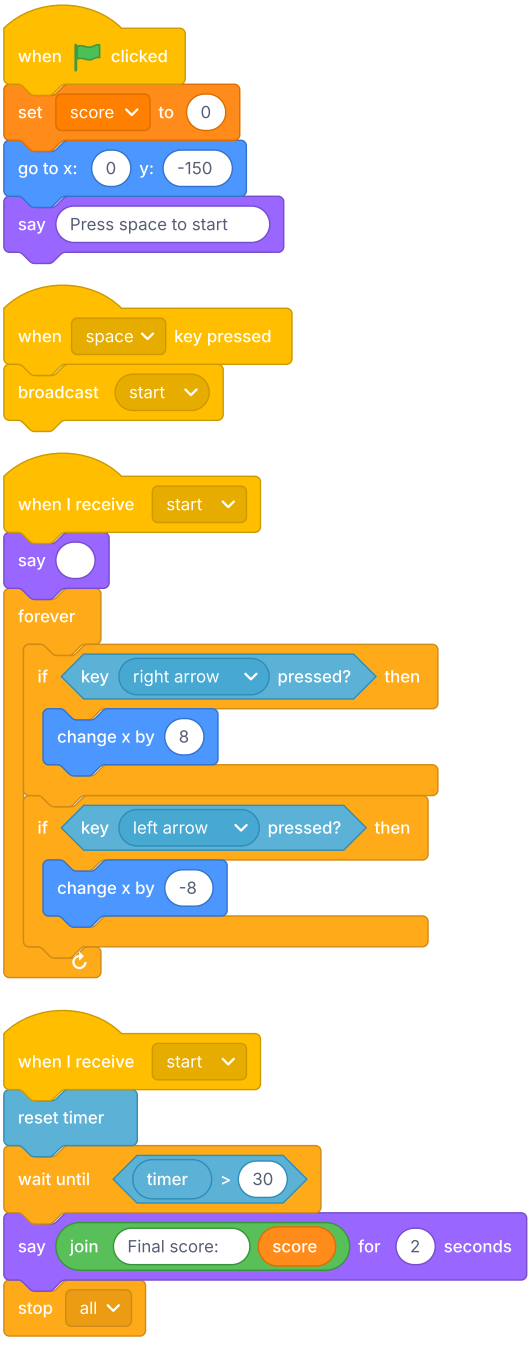
Clones: Fifty falling raindrops, Ten enemies that each move by themselves
List: Remembering every word the player typed
My Block: Drawing the same star shape in many places, Code repeated in three scripts
Broadcast: Starting the game when space is pressed

2. Capstone: Arcade Engineer (probe · code lab)

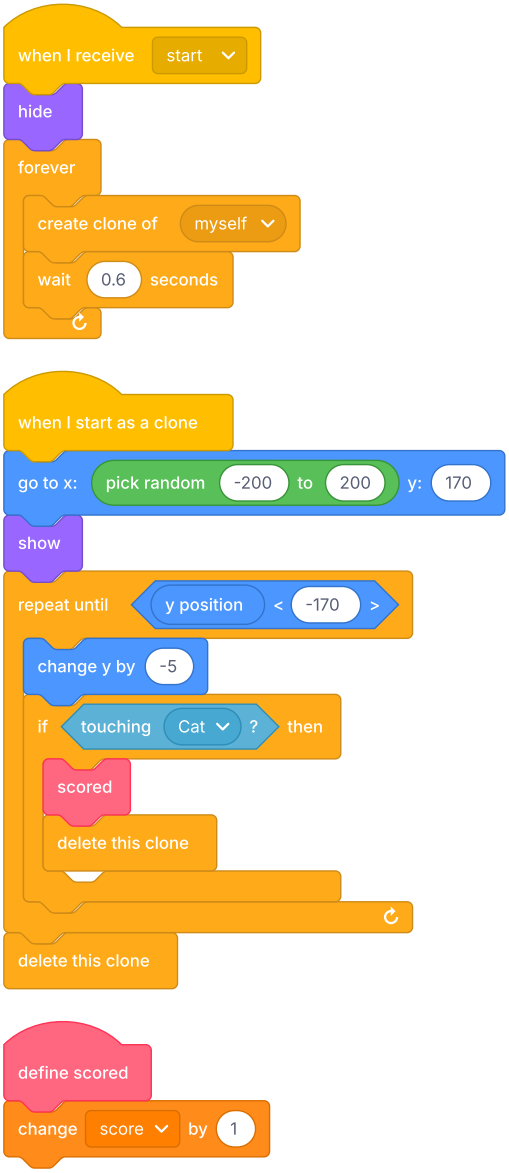
Goals checked: 2+ sprites · Start state · Clones or a list · A custom block · Score starts at 0 · An ending

Example project:

Cat



Fish



Answer key — worksheet

1. **Key words:** game design = A plan of how your game works. · playtest = Watching others play to find problems. · iteration = Improving in small cycles.
2. **Which tool would you use?:** Clones: Fifty falling raindrops, Ten enemies that each move by themselves · List: Remembering every word the player typed · My Block: Drawing the same star shape in many places, Code repeated in three scripts · Broadcast: Starting the game when space is pressed
3. **My plan:** A clear plan that covers every requirement.
4. **Checklist:** All requirements met.
5. **Two stars and a wish:** Kind, specific feedback and an improvement.

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.